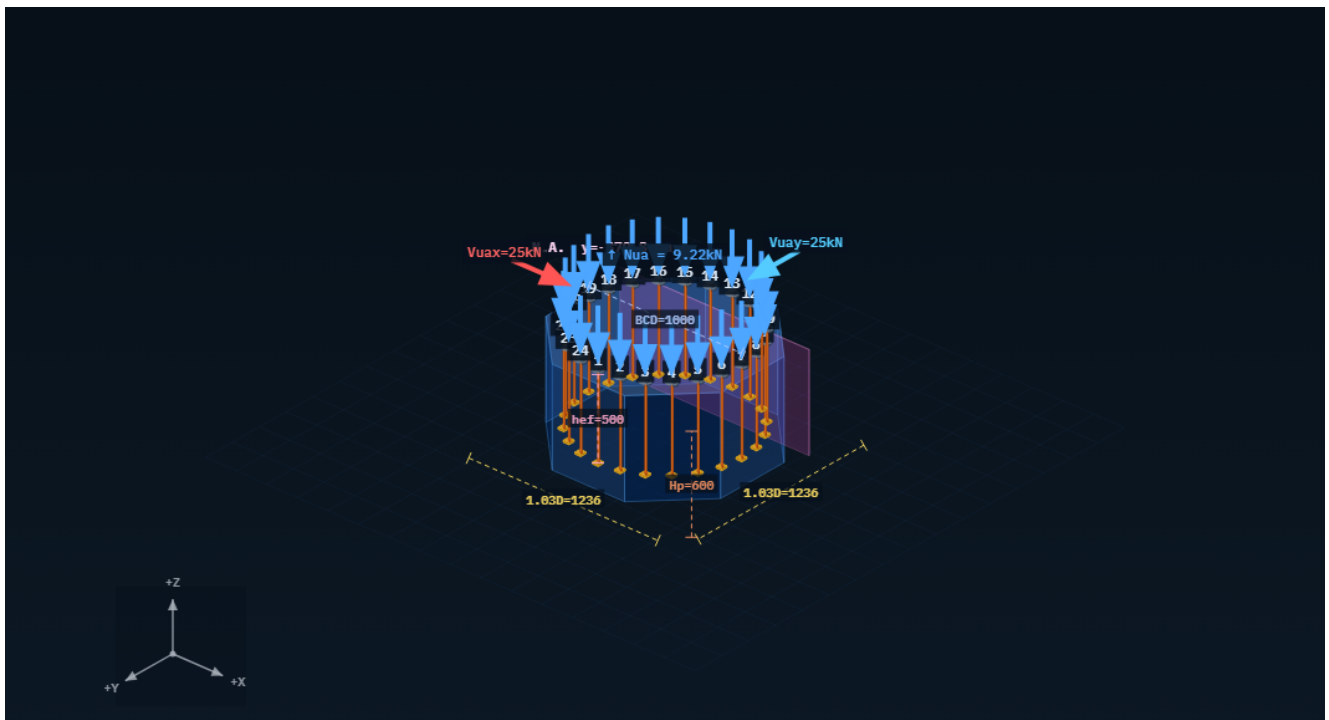


Anchor Bolt Structural Design

ACI 318-19 Chapter 17 — Anchoring to Concrete

Job No.: —
Project: —
Doc. No.: —
Item: —
Engineer: —
Date: 2026-08-03

3D Model View — Isometric View



Result Summary					
Failure Mode	Symbol	Capacity	Demand	Ratio	Status
T1 Steel (Tension)	ϕN_{sa}	145.05 kN	9.22 kN	6.40%	O.K
T2 Breakout (Tension)	ϕN_{cbg}	332.70 kN	92.39 kN	27.80%	O.K
T3 Pullout	ϕN_{pn}	148.34 kN	9.22 kN	6.20%	O.K
T4 Side - face	ϕN_{sb}	103.50 kN	92.39 kN	89.30%	O.K
V1 Steel (Shear)	ϕV_{sa}	93.12 kN	1.47 kN	1.60%	O.K
V2 Breakout (Shear)	ϕV_{cbg}	1797.63 kN	1.47 kN	0.10%	O.K
V3 Pryout	ϕV_{cpg}	665.40 kN	1.47 kN	0.20%	O.K
Combined T + V	IR	0.89	1	89.30%	O.K

1. DESIGN INPUT CONDITIONS

Parameter	Symbol	Value / Description	Unit
Anchor Type	—	Cast - in Headed Bolt	
No. of Anchors (nx ny)	24 × 1	24.00 ea — GROUP	ea
Bolt Diameter	da	20.00	mm
Effective Embed. Depth	hef	500.00	mm
Neutral Axis Method	N.A.	Linear Stress Method	
Linear Stress Assumption	C / T	Concrete compression only + anchor tension only; c solved from C = T + P and k from moment equilibrium	
Modified hef	hef'	78.67 (Modified)	mm
hef' Calc.	—	= R s _{X,max} Group check: s _{X,max} < 3hef 130.5 < 1500.0 Group edge count (< 1.5hef) = 4 ca,max = 118.0 mm s _{X,max} = 130.5 mm hef' = max (ca,max / 1.5, s _{X,max} / 3) = max (78.7, 43.5) = 78.7 mm	
Head Width (AC)	eh	34.60	mm
Bearing Area	Abrg	883.0 = eh ² - ϕ / 4 × da ²	mm ²
futa (limited)	futa	min (860, 1.9 × 400, 860) = 760 MPa	MPa
fya	fya	400.00	MPa
Steel Element Behavior	—	Ductile steel element governs	Table 17.5.3 (a)
Pedestal Lpx	Lpx	1236.00	mm
Pedestal Lpy	Lpy	1236.00	mm
Pedestal Height	Hp	600.00	mm
Concrete Strength	f'c	30.00	MPa
Lightweight Factor	λ	1.00	—
Crack Condition	—	Cracked	
Edge Distance X	Ca1,x	(1236 - 1000) / 2 = 118 mm	mm
Edge Distance Y	Ca1,y	(1236 - 1000) / 2 = 118 mm	mm
Governing Ca1	Ca1	min (118, 118) = 118 mm	mm

Design Tension	Nua	9.22	kN
Design Shear X	Vua,x	25.00	kN
Design Shear Y	Vua,y	25.00	kN
Resultant Shear	Vua	35.35 kN	kN
Rebar Dia.	dv	None	
Total Rebar	Ntot	0.00	ea
N1 (within 0.5hef)	N1	0.00	ea
0.5hef Zone	0.5hef	250.00	mm
Rebar fy	fy	400.00	MPa
Cover C1	C1	50.00	mm
Cover C2	C2	75.00	mm
Tie Bar	dtie	D16	
ϕ Steel Tension	ϕs,t	0.75 (ductile)	Table 17.5.3 (a)
ϕ Steel Shear	ϕs,v	0.65 (ductile)	Table 17.5.3 (a)
ϕ Conc. Tension	ϕc,t	0.70	Table 17.5.3 (b)
ϕ Conc. Shear	ϕc,v	0.70	Table 17.5.3 (b)

2. TENSILE STRENGTH — ACI 318-19 §17.6

T1. Steel Failure in Tension [ϕNsa] — §17.7.1.2

Item	Sym.	Formula / Calculation	Value	Ref.
Bolt dia. 0.9da	0.9da	$0.9 \times 20 = 18.0 \text{ mm}$	18.00 mm	§17.6.1.2
Eff. tensile area Ase,N	Ase,N	$\phi / 4 \times (0.9da)^2 = \phi / 4 \times 18.0^2$	254.47 mm²	§17.6.1.2
Limited futa	futa	$\min (860, 1.9 \times 400, 860) = 760 \text{ MPa}$	760.00 MPa	§17.6.1.2
Nom. strength (1ea)	Nsa,1	$Ase,N \times futa = 254.47 \times 760 / 1000$	193.40 kN	§17.7.1.2
ϕ steel tension	ϕs,t	Ductile steel element governs	0.75	Table 17.5.3 (a)
ϕNsa (governing)	ϕNsa	$\phi \times n \times Ase,N \times futa = 0.75 \times 24 \times 254 \times 760 / 1000$	145.05 kN	§17.7.1.2

[O.K] ϕNsa Nua: 145.05 kN 9.22 kN (6.4%)

T2. Concrete Breakout in Tension [φNcbg] — §17.6.2.1b				
Item	Sym.	Formula / Calculation	Value	Ref.
Projected Area ANc				
ANc0	ANc0	$9 \times he_f^2 = 9 \times 78.7^2$	55696.00 mm²	§17.6.2.1.4
ANc Width	ANc_W	$\min (1.5h, Ca1x) + sx + \min (1.5h, Ca1x) = 118 + 1000 + 118$	1236.00 mm	§17.6.2.1.2
ANc Height	ANc_H	$\min (1.5h, Ca1y) + sy + \min (1.5h, Ca1y) = 118 + 1000 + 118$	1236.00 mm	§17.6.2.1.2
ANc	ANc	$ANcW \times ANcH = 1236 \times 1236$	1527696.00 mm²	§17.6.2.1.2
ANc / ANc0	ratio	1527696 / 55696	27.43	
Basic Concrete Breakout Strength Nb				
kc	kc	cast - in = 10 / post - installed = 7	10.00	§17.6.2.2.2
Lambda	λ	Lightweight factor	1.00	§19.2.4
f'c	f'c	30	5.48	
hef^1.5	hef^1.5	78.7^1.5	697.73	
Nb	Nb	$kc \times \lambda a \times f'c \times hef^{1.5} = 10 \times 1 \times 5.4772 \times 697.73 / 1000$	38.22 kN	§17.6.2.2.1
Modification Factors				
ψec,N	ψec,N	No eccentricity 1.0	1.00	§17.6.2.3
ψed,N	ψed,N	Ca1 1.5hef 1.0	1.00	§17.6.2.4
ψc,N	ψc,N	Cracked 1.0	1.00	§17.6.2.5
ψcp,N	ψcp,N	Cast - in 1.0	1.00	§17.6.2.6
Breakout Strength				
Ncbg	Ncbg	$(ANc / ANc0) \times \psi_{ec} \times \psi_{ed} \times \psi_c \times \psi_{cp} \times Nb$	475.28 kN	§17.6.2.1b
φc,t	φc,t	Supplementary Reinforcement: No	0.70	Table 17.5.3 (b)
φNcbg	φNcbg	0.7×475.2826	332.70 kN	§17.6.2.1b
[O.K] φNcbg Nua: 332.70 kN 9.22 kN (27.8%)				
T3. Pullout Failure [φNpn] — §17.6.3.1				
Item	Sym.	Formula / Calculation	Value	Ref.
Bearing Area	Abrg	$eh^2 - \phi / 4 \times da^2 = 34.6^2 - \phi / 4 \times 20^2$	883.00 mm²	§17.6.3.2.2

Np (1 anchor)	Np	$8 \times \text{Abrg} \times f_c = 8 \times 883 \times 30 / 1000$	211.92 kN	\$17.6.3.2.2
ψ_c, P	ψ_c, P	Cracked 1.0	1.00	\$17.6.3.3
ϕN_{pn}	ϕN_{pn}	$\phi \times \psi_c, P \times N_p \times n = 0.7 \times 1 \times 211.9200 \times 24$	148.34 kN	\$17.6.3.1

[O.K] ϕN_{pn} Nua: 148.34 kN 9.22 kN (6.2%)

T4. Side-face Blowout [ϕN_{sb}] — §17.6.4.1

Item	Sym.	Formula / Calculation	Value	Ref.
Applicability	—	$Ca1 < 0.4h_{ef}$ 118 < 200 mm	APPLIES	\$17.6.4.1
Nsb	Nsb	$160 \times Ca1 \times \text{Abrg} \times \lambda \times f_c$	147.85 kN	\$17.6.4.1
ϕN_{sb}	ϕN_{sb}	0.7×147.8495	103.50 kN	\$17.5.3

[O.K] ϕN_{sb} Nua: 103.49 kN 9.22 kN (8.9%)

3. SHEAR STRENGTH — ACI 318-19 §17.7

Resultant Shear Force

Item	Sym.	Formula	Value	Ref.
Shear X	Vua,x	Input	25.00 kN	input
Shear Y	Vua,y	Input	25.00 kN	input
Resultant	Vua	$(Vu_{ax}^2 + Vu_{ay}^2)^{0.5} = (25^2 + 25^2)^{0.5}$	35.35 kN	\$17.7

V1. Steel Failure in Shear [ϕV_{sa}] — §17.7.1.2b

Item	Sym.	Formula / Calculation	Value	Ref.
Eff. shear area Ase,V	Ase,V	$\phi / 4 \times da^2 = \phi / 4 \times 20^2$	314.16 mm ²	\$17.7.1.2
futa (limited)	futa	760	760.00 MPa	\$17.6.1.2
ϕ steel shear	ϕ_s, v	Ductile steel element governs	0.65	Table 17.5.3 (a)
ϕV_{sa}	ϕV_{sa}	$\phi \times 0.6 \times A_{se,V} \times f_{uta} \times n = 0.65 \times 0.6 \times 314 \times 760 \times 24 / 1000$	93.12 kN	\$17.7.1.2b

[O.K] ϕV_{sa} Vua: 93.12 kN 35.36 kN (1.6%)

V2. Concrete Breakout in Shear [ϕV_{cbg}] — §17.7.2.1b

Item	Sym.	Formula / Calculation	Value	Ref.
Projected Area AVc				
AVc0	AVc0	$4.5 \times Ca1^2 = 4.5 \times 118^2$	62658.00 mm ²	\$17.7.2.1.3

Height hc	hc	$\min (H_p, 1.5Ca_1) = \min (600, 177)$	177.00 mm	\$17.7.2.1.2
Width	—	$\min (1.5Ca_1, Ca_1y) + s_y + \min (1.5Ca_1, Ca_1y)$	0.00 mm	\$17.7.2.1.2
AVc	AVc	$AVc = \min (1.5ca_1D, n \times 4.5ca_1^2) = \min (212400.0, 1503792.0) = 212400.0 \text{ mm}^2$	212400.00 mm ²	
AVc / AVc0	ratio	212400 / 62658	3.39	
Basic Shear Strength Vb				
le	le	$\min (8 \times da, hef) = \min (160, 79)$	160.00 mm	\$17.7.2.2.1
Vb1	Vb1	$7 \times (le / da)^{0.2} \times da \times \lambda \times f_c \times Ca_1^{1.5}$	832.74 kN	\$17.7.2.2.1
Vb2	Vb2	$9 \times \lambda \times f_c \times Ca_1^{1.5}$	757.57 kN	\$17.7.2.2.1
Vb	Vb	$\min (Vb_1, Vb_2)$	757.57 kN	—
Modification Factors				
$\psi_{ec,V}$	$\psi_{ec,V}$	No eccentricity	1.00	\$17.7.2.3
$\psi_{ed,V}$	$\psi_{ed,V}$	Ca2 1.5Ca1 1.0	1.00	\$17.7.2.4
ψ_c,V	ψ_c,V	Cracked 1.0	1.00	\$17.7.2.6
ψ_h,V	ψ_h,V	Hp 1.5Ca1 1.0	1.00	\$17.7.2.5
Breakout Shear Strength				
Vcbg	Vcbg	$(AVc / AVc_0) \times \psi_{ec} \times \psi_{ed} \times \psi_c \times \psi_h \times Vb$	2568.05 kN	\$17.7.2.1b
ϕ_c,v	ϕ_c,v	Supplementary Reinforcement: No	0.70	Table 17.5.3 (b)
$\phi Vcbg$	$\phi Vcbg$	0.7×2568.0474	1797.63 kN	\$17.7.2.1b
[O.K] $\phi Vcbg$ V_{ua} : 1797.63 kN 35.36 kN (2.0%)				
V3. Pryout [$\phi Vcpg$] — §17.7.3.1				
Item	Sym.	Formula / Calculation	Value	Ref.
kcp	kcp	hef 65mm 2	2.00	\$17.7.3.1
Ncbg	Ncbg	(from T2)	475.28 kN	T2
Vcpg	Vcpg	$kcp \times Ncbg = 2 \times 475.2826$	950.57 kN	\$17.7.3.1
$\phi Vcpg$	$\phi Vcpg$	0.7×950.5652	665.40 kN	\$17.5.3
[O.K] $\phi Vcpg$ V_{ua} : 665.40 kN 35.36 kN (5.3%)				

4. COMBINED TENSION + SHEAR — ACI 318-19 §17.8

Item	Sym.	Formula / Calculation	Value	Ref.
ϕN_n (governing)	ϕN_n	$\min (\phi N_{sa}, \phi N_{cbg}, \phi N_{pn}, \phi N_{sb})$	103.50 kN	§17.8.1
ϕV_n (governing)	ϕV_n	$\min (\phi V_{sa}, \phi V_{cbg}, \phi V_{cpg})$	93.12 kN	§17.8.1
Tension Ratio	$N_u / \phi N_n$	$9.221 \sim 92.393 / 103.495$	0.89	§17.8.3
Shear Ratio	$V_u / \phi V_n$	$1.473 \sim 35.355 / 93.117$	0.38	§17.8.3
Interaction Mode	—	—		§17.8.3
Interaction Ratio	IR	Single dominant	0.89 1	§17.8.3
[O.K] IR 1: 0.8927 1 (89.3%)				